

DAY 08 — PYTHON LOOPS

90 Days Python Challenge • Notes + Examples + Practice

1. What is a Loop?

A **loop** repeats a block of code multiple times. Loops help us avoid writing the same code again and again.

Python has two main loops: **for loop** and **while loop**.

2. for Loop

A **for loop** is used to iterate over a sequence, such as a list, string, or range of numbers.

Syntax

```
for variable in sequence:
    statements
```

Example 1 — Print numbers from 1 to 5

```
for i in range(1, 6):
    print(i)
```

Output: 1 2 3 4 5

Remember: range(1, 6) starts at 1 and stops before 6, so it produces 1 through 5.

Example 2 — Loop through a string

```
for letter in "PYTHON":
    print(letter)
```

3. while Loop

A **while loop** repeats as long as its condition is **True**. Update the loop variable so the condition eventually becomes False.

Syntax

```
while condition:
    statements
```

Example — Print numbers from 1 to 5

```
i = 1
while i <= 5:
    print(i)
    i += 1
```

Output: 1 2 3 4 5

4. for Loop vs while Loop

for loop	while loop
Useful when iterating over a sequence or a known range.	Useful when repetition depends on a condition.
Often used with lists, strings, and range().	Needs a condition that can become False.
Usually manages sequence iteration for you.	Remember to update variables to avoid an infinite loop.

5. Important Loop Controls

break — stops the loop immediately.

continue — skips the current iteration and moves to the next one.

pass — does nothing; used as a placeholder.

Example — break

```
for i in range(1, 6):
    if i == 4:
```

```
        break
    print(i)
```

Output: 1 2 3

6. Common Mistakes

- Forgetting the colon (:) after the loop statement.
- Incorrect indentation inside the loop.
- Forgetting to update the variable in a while loop, causing an infinite loop.
- Assuming `range(1, 5)` includes 5 — it does not.

7. Practice Questions — Day 08

1. Print numbers from 1 to 10 using a for loop.
2. Print all even numbers from 2 to 20.
3. Print each character of your name on a new line.
4. Find the sum of numbers from 1 to 10 using a loop.
5. Print numbers from 10 down to 1 using a while loop.
6. Ask the user for a number and print its multiplication table.
7. Use `break` to stop a loop when the number reaches 6.
8. Use `continue` to skip number 5 while printing 1 to 10.

One-minute revision: `for` = iterate over a sequence/range • `while` = repeat while condition is True • `break` = stop • `continue` = skip current iteration • `pass` = placeholder